

Network Design and Implementation for Buildings F, G, and Workshops

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Abstract—

Computer networks have become a fundamental component of modern educational institutions, as they enable communication, information sharing, and access to digital resources. Universities rely heavily on computer networks to support e-learning platforms, administrative systems, research laboratories, and workshop environments. As the number of users and connected devices increases, the need for a well-structured, secure, and scalable network infrastructure becomes essential.

This project focuses on the design and implementation of a campus network that connects Building F, Building G, and the Workshops Building at New Ismailia National University. The proposed network aims to provide reliable connectivity, efficient traffic management, and strong security mechanisms. The design follows real-world networking principles commonly used in enterprise and campus networks and is validated through simulation using Cisco Packet Tracer.

Keywords—

Computer Networks, VLAN, IP Addressing, Network Security, Campus Network

I. Introduction

Modern university campuses require reliable and secure computer networks to support academic, administrative, and laboratory activities. This project proposes a professional campus network design connecting Building F, Building G, and the Workshops Building.

II. Network Requirements Analysis

The proposed network is designed to serve different categories of users, each with specific requirements. Students require stable internet access, connectivity to laboratory computers, and access to learning management systems. Academic staff depend on the network for research activities, communication, and access to shared resources. Administrative staff use the network for databases, management systems, and official communications. In addition, network administrators require secure access to manage and monitor network devices.

Based on the expected number of users, Building F supports approximately 250 users, Building G supports around 300 users, and the Workshops Building supports nearly 150 users. The network must therefore support high bandwidth, low latency, and continuous availability. These requirements influenced the selection of network topology, hardware devices, and addressing schemes.

III. Network Topology Design

A hierarchical network topology is adopted in this project to ensure scalability, reliability, and ease of management. The topology consists of three layers: the core layer, the distribution layer, and the access layer. The core layer is responsible for routing traffic between the three buildings and acts as the backbone of the network. The distribution layer aggregates traffic from each building and applies network policies such as VLANs and access control. The access layer connects end-user devices, including computers and printers, to the network.

As shown in Fig. 1, each building contains multiple access switches connected to a distribution switch. All buildings are interconnected through a central router, allowing efficient inter-building communication. This design reduces broadcast traffic, improves fault isolation, and simplifies troubleshooting.

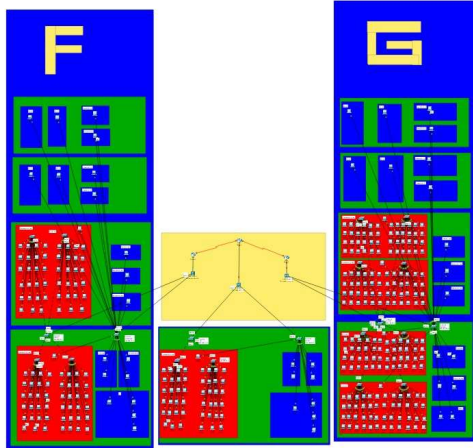


Fig. 1. Network topology for Buildings F, G, and Workshops

IV. IP Addressing and VLAN Design

To efficiently manage network resources, a private IP addressing scheme is implemented using structured subnetting. VLANs are used to logically segment the network based on user roles, such as students, academic staff, administrative staff, and network management. This segmentation improves performance by reducing unnecessary broadcast traffic and enhances security by isolating sensitive data.

Inter-VLAN routing is configured using a router or Layer 3 switch, allowing controlled communication between VLANs. This approach ensures that only authorized traffic can flow between different network segments.

V. Hardware and Equipment Selection

The selection of networking hardware is based on reliability, scalability, and compatibility with the simulation environment. Cisco routers are used for inter-building routing due to their robust routing capabilities. Cisco Catalyst switches are deployed at the distribution and access layers to support VLANs, trunking, and port security. These devices are widely used in enterprise networks and provide features required for a professional campus network design.

VI. Network Security Architecture

Security is a critical aspect of the proposed network. VLAN-based segmentation provides the first layer of protection by isolating different user groups. Access Control Lists (ACLs) are implemented to restrict unauthorized access between VLANs and critical network resources. Port security is enabled on access switches to prevent unauthorized devices from connecting to the network. Management access is restricted to authorized administrators only, ensuring secure network operation.

VII. Scalability and Future Expansion

The proposed network design supports future expansion without major changes to the existing infrastructure. Additional switches and VLANs can be added as the number of users increases. New buildings can be connected to the core layer, and advanced services such as servers or wireless networks can be integrated easily. This makes the network suitable for long-term university growth.

VIII. Simulation and Implementation

The complete network design was implemented using Cisco Packet Tracer. All devices were configured with appropriate IP addresses, VLANs, trunk links, and routing settings. Connectivity was tested using ping commands to verify communication between devices within the same building and across different buildings. The successful simulation confirms the correctness and feasibility of the proposed design.

IX. Conclusion

This project presented a detailed and professional design of a campus computer network connecting Building F, Building G, and the Workshops Building. The proposed solution achieves secure, scalable, and efficient connectivity suitable for a university environment. The use of hierarchical topology, VLAN segmentation, and proper security mechanisms demonstrates a real-world approach to network design.

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